

KERALA AGRICULTURAL UNIVERSITY[Home](#)[Institution](#)[Education](#)[Research](#)[Extension](#)[News](#)[Contact Us](#)[Back to Contents](#)**CEREALS AND MILLETS****RICE (*Oryza sativa*)**

Rice can be cultivated under a variety of climatic and soil conditions. Rice cultivation is conditioned by temperature parameters at the different phases of growth. The critical mean temperature for flowering and fertilization ranges from 16 to 20°C, whereas, during ripening, the range is from 18 to 32°C. Temperature beyond 35°C affects grain filling. Rice comes up well in different soil types. For normal growth, a pH range of 5.0-8.0 is suitable. In general, rice can be grown as transplanted or direct sown crop during three seasons as shown below depending on the agroclimatic situations



During second crop, for higher yield in photosensitive high yielding varieties in Palakkad district, where assured irrigation is available, the crop commencement may be adjusted in such a way that it flowers only during the second fortnight of December, facilitating proper integration with the weather, better utilization of applied fertilizers and high filling percentage.

Table 2. Rice varieties suited for different situations

Region/Type of land	Seasons	Varieties
Upland (Modan land) Purely rainfed	First crop	PTB 28, PTB 29, PTB 30, Suvarnamodan, Annapoorna, Matta Triveni, Swarnaprabha, Rohini, Aiswarya, Vaishak, Harsha
Palliyals (Myals) Single crop terraced	First crop	Short duration : Rohini, Annapoorna, Matta Triveni, Jyothi, Kairali, Kanchana, Harsha, Karthika, Ahalya, Prathyasa Medium duration : Aswathy, Sabari, Bharathy, Jaya, Aiswarya, Aathira. Long duration : Mahsuri
Double crop wetlands : a. Semi-dry cultivation	First crop	Short duration : Matta Triveni, Annapoorna, Jyothi, Aruna, Makom, Swarnaprabha, Ahalya, Varsha, Rohini, Karthika, Revathi, Remanika, Krishnanjana, Kanchana, Harsha, Kairali, Kunjukunju Varna, Kunjukunju Priya, Prathyasa Medium duration : Aswathy, Sabari, Bharathy, Jaya, Arathy, Remya, Kanakom, Ranjini, Pavithra, Panchami, Uma, Aathira, Aiswarya, Pavizham, Bhadra Long duration : Mahsuri
	Second crop	Any of the varieties suggested for the first crop season (Except Rohini)
b. Transplanted	First crop	Short duration : Annapoorna, Matta Triveni, Jyothi, Swarnaprabha, Kairali, Kanchana, Karthika, Aruna, Makom, Revathi, Remanika, Krishnanjana, Varsha, Rohini, Ahalya, Kunjukunju Varna, Kunjukunju Priya, Prathyasa Medium duration : Jaya, Sabari, Bharathy, Aswathy, Aathira, Aiswarya, Pavizham, Remya, Kanakom, Ranjini, Pavithra, Panchami, Uma Long duration : Mangala Mahsuri, Pranava, Mahsuri
	Second crop	Short duration : Annapoorna, Matta Triveni, Jyothi, Kairali, Kanchana, Karthika, Makom, Revathi, Remanika, Krishnanjana, Kunjukunju Varna, Kunjukunju Priya, Prathyasa Medium duration : Jaya, Sabari, Bharathy, Aswathy, Aathira, Aiswarya, Pavizham, Remya, Kanakom, Ranjini, Pavithra, Panchami, Uma, Karishma Long duration : Mangala Mahsuri, Karuna, Rasmi, Nila, Makaram, Kumbham, Dhanu, Anashwara, Mahsuri
	Third crop	Short duration : Annapoorna, Matta Triveni, Jyothi, Aruna, Swarnaprabha, Kairali, Kanchana, Karthika, Makom, Revathi, Remanika, Krishnanjana, Ahalya, Harsha, Varsha, Prathyasa Medium duration : Jaya, Sabari, Bharathy, Aathira, Aiswarya, Pavizham, Remya, Kanakom, Ranjini, Pavithra, Panchami, Uma
Kuttanad area	Puncha	Short duration : Karthika, Makom, Jyothi, Matta Triveni, Annapoorna, Revathi, Remanika, Krishnanjana, Prathyasa Medium duration : Kanakom, Jaya, Sabari, Bharathy, Ranjini, Pavithra, Panchami, Uma, Gouri, Bhadra, Asha, Pavizham

	Additional crop	Short duration : Karthika, Makom, Aruna, Jyothi, Matta Triveni, Annapoorna, Revathi, Remanika, Krishnanjana, Prathyasa Medium duration : Pavizham, Remya, Kanakom, Jaya, Sabari, Ranjini, Pavithra, Panchami, Uma
Kole area	Mundakan	Extra short duration : Hraswa Short duration : Karthika, Makom, Aruna, Jyothi, Onam, Bhagya, Matta Triveni, Annapoorna, Revathi, Remanika, Krishnanjana, Ahalya, Varsha, Swarnaprabha, Kanchana, Kairali, Manupriya, Prathyasa Medium duration : Pavizham, Remya, Kanakom, Jaya, Ranjini, Pavithra, Panchami, Uma, Bhadra, Aswathy, Sabari, Bharathy and Aiswarya
Pokkali area	First crop	Vyttila-1, Vyttila-2, Vyttila-3, Vyttila-4, Vyttila-5, Vyttila-6, Vyttila-7, Vyttila-8
Kaippad tract	First crop	Ezhome-1, Ezhome-2
Deep ill-drained regions of southern districts	First crop	Remya, Arathi, Uma
	Second crop	Kottarakkara-1, Lakshmi, Nila, Makaram, Kumbham, Mangala Mahsuri
Waterlogged and flooded areas	First crop	IR-5, Pankaj, Jagannath, H4, Mahsuri, Neeraja, Mangala Mahsuri
Oorumundakan	Second crop	Late duration : Sagara
Onattukara and coastal sandy areas a. Where HYVs do not come up	First crop	PTB 23
	Second crop	PTB 20
b. Where HYVs come up well	First crop	Short duration : Annapoorna, Matta Triveni, Jyothi, Chingam, Bhagya, Rohini, Onam, Aruna, Makom, Karthika, Revathi, Remanika, Krishnanjana, Prathyasa. Medium duration : Jaya, Sabari, Bharathy, Aswathy, Pavizham, Remya, Kanakom, Arathi, Renjini, Pavithra, Panchami, Uma
	Second crop	Short duration : Annapoorna, Matta Triveni, Jyothi, Makom, Karthika, Revathi, Remanika, Krishnanjana Medium duration : Jaya, Sabari, Bharathy, Aswathy, Bhadra, Asha, Pavizham, Remya, Kanakom, Dhanya, Anaswara, Dhanu (season bound), Ranjini, Pavithra, Panchami, Uma
	Third crop	Short duration : Annapoorna, Matta Triveni, Rohini, Makom, Revathi, Remanika, Krishnanjana, Prathyasa. Medium duration : Jaya, Sabari, Bharathy, Aswathy, Pavizham, Remya, Kanakom, Ranjini, Pavithra, Panchami, Uma, Karishma
Poonthalpadam		Neeraja, Swetha, Ponmani
High altitude area : a. Single crop areas		WND-1, WND-2, Aswathy, Jaya, Sabari, Mahsuri, Bhadra, IR 8, Aathira, Uma, Deepthi
b. Double crop areas	First crop	Aswathy, Jaya, Sabari, Bharathy, Bhadra, Deepthi, Aathira, IR-8, Uma
	Second crop	Aswathy, Jaya, Sabari, Bharathy, Bhadra, Deepthi, Aathira, IR-8, Uma
c. Eastern lateritic regions of Kollam and Alappuzha districts	Second crop	Lakshmi, Makaram, Kumbham, Thulam
Koottumundakan	First crop Second crop	Samyuktha, Swarnapraha, Vaishak, Karthika, Aiswaraya Makaram, Kumbham
Chittoor black soil	First crop	ASD 16, ASD 17, Mahsuri, Varsha, ADT 43, Ranjini
	Second crop	Ponni, Vellaponni, Ponmani, ASD 16, ASD 17, Pranava, Karuna, Swetha, Bhadra

Table 3. Important characteristics of varieties recommended for cultivation

Sl No.	Variety (days)	Duration	Bran colour, grain type	Characteristics
Extra-short duration				
1	Hraswa (Cul 24-20)	75-80	Red, medium bold	Ideal as a contingent variety for areas where there is crop loss. Susceptible to leaf folder. Raised only as direct sown crop
Short duration				
1	Kattamodan (PTB 28)	110-115	Red, long bold	For uplands (modan). Tolerant to drought
2	Karutha Modan (PTB 29)	105-110	Red, long bold	-do- Tall
3	Chuvanna Modan (PTB 30)	105-110	Red, long bold	-do- Tall
4	Annapoorna (PTB 35)	95-100	Red, short bold	Suitable for direct seeding. Susceptible to blast, sheath blight and BPH. Suited for I and III crop seasons
5	Rohini (PTB 36)	85-105	White, long bold	Performs well during Virippu season. Not recommended for <i>mundakan</i> season. Suitable for direct seeding
6	Triveni (PTB 38)	100-105	White, long bold	Tolerant to BPH. Susceptible to blast and sheath blight
7	Jyothi (PTB 39)	110-115	Red, long bold	Moderately tolerant to BPH and blast; susceptible to sheath blight; suitable for direct seeding, transplanting and special systems of Kole and Kuttanad
8	Swarnaprabha (PTB 43)	105-110	White, long bold	Suited for upland (modan) and for all the three seasons in the wet-lands. Susceptible to sheath blight and bacterial blight. Moderately resistant to stem borer. A good first crop component in <i>Kootumundakan</i>
9	Matta Triveni (PTB 45)	100-105	Red, medium bold	Tolerant to BPH, susceptible to blast and sheath blight. Suitable for I and III crop
10	Kairali (PTB 49)	110-115	Red, long bold	Moderately resistant to blast, blight, gall midge and leaf folder. Can be cultivated in all the three seasons

11	Kanchana (PTB 50)	105-110	Red, long bold	Suitable for <i>Kole and Kuttanad</i> regions. Resistant to blight, blast, stem borer and gall midge. Suitable for all seasons
12	Karthika (MO 7)	105-110	Red, long bold	Suitable for growing in all the three seasons. Moderate resistance to sheath blight, sheath rot and BPH. A good first crop component in <i>Kootumundakan</i>
13	Aruna (MO 8)	100-110	Red, medium bold	Tolerant to BPH and stem borer, moderately resistant to gall midge, sheath rot. Dormancy up to one month. Specifically suited to wet season
14	Makom (MO 9)	100-110	Red, short bold	Moderately resistant to pests like BPH, stem borer, gall midge, leaf folder and diseases like sheath blight and sheath rot. Dormancy up to one month. Specifically suited for wet season. Can be cultivated in all the three seasons
15	Remanika (MO 15)	100-105	Red, short bold	Dwarf, medium tillering, resistant to BPH and moderately resistant to gall midge
16	Revathi (MO 17)	105-110	Red, medium bold	Dwarf, medium tillering, resistant to BPH and moderately resistant to gall midge. Dormancy upto 3 weeks. Suited to all the three seasons especially to additional crop of Kuttanad
17	Krishnanjana (MO 19)	105-110	Red, medium bold	Medium tillering, resistant to BPH and dormancy upto 3 weeks, suited to all seasons especially to kari lands of Kuttanad, tolerant to iron toxicity
18	Bhagya (Kayamkulam 2)	100	Red	Suitable for first crop season in Onattukara and eastern lateritic region. Drought resistant in the early growth phases. Moderately tolerant to blight, stem borer, leaf roller and case worm
19	Onam (Kayamkulam 3)	95	Red	Drought tolerant in the early growth phases. Moderately resistant to blight, sheath blight and blast. Suited for dry sowing in the first crop season of Onattukara region
20	ASD 17	100-105	White, short bold	Good grain quality and high yielding

21	Ahalya	90-100	Red	Good cooking quality, tolerant to leaf folder and moisture stress in the early growth phase
22	Harsha (PTB 55)	105-110	Red, long bold	Suitable for direct seeding in rainfed lowlands. Moderate resistance to blast and moisture stress
23	Varsha (PTB 56)	110-115	Red, long bold	Suitable for direct seeding and transplanting in irrigated lowlands. Moderate resistance to blue beetle
24	Kunjukunju Vama (VK-1)	110-115	Red, short bold	Photo-insensitive, moderately tolerant to major pests like gall fly, leaf folder, whorl maggot and stem borer. Adapted for double cropped areas of Palakkad, Thrissur and Ernakulam districts as direct seeded or transplanted crop. Leaf sheath and apicules are purple pigmented
25	Kunjukunju Priya (VK-2)	105-110	Red, short bold	Photo-insensitive, moderately tolerant to major pests like gall fly, leaf folder, whorl maggot and stem borer. Adapted for double cropped areas of Palakkad, Thrissur and Ernakulam districts as direct seeded or transplanted crop. This is a non-pigmented variety
26	Chingam	95-100	Red, short bold	Photo-insensitive semi-tall variety suitable for first crop season in Onattukara. Moderately resistant to sheath blight and brown leaf spot
27	Manupriya	105-110	Red, long bold	Suitable for Kole lands. Tolerant to sheath blight, brown spot, blast, stem borer and gall midge. Suitable for all seasons
28	Prathyasa (MO-21)	100-110	Red, long bold	Non-lodging, Photoinsensitive, semi-tall variety, suitable for Kuttanad. Moderately resistant to gall midge. BPH, sheath blight and sheath rot
29	Samyuktha (PTB 59)	112-117	Red, short bold	Suitable for Koottumundakan system of cultivation with Makaram in Central zone
30	ASD 16	110-115	White, short bold	Good grain quality, high yield

Medium duration				
1	Jaya	120-125	White, long bold	Very high yield potential, highly susceptible to BPH, and other pests
2	Aswathy (PTB 37)	120-125	White, long bold	Suitable for dry sowing in the first crop season
3	Sabari (PTB 40)	130-135	Red, long bold	Susceptible to sheath blight
4	Bharathy (PTB 41)	120-125	Red, long bold	Tolerant to BPH, moderately resistant to blast, suitable for dry sowing
5	Suvarnamodan (PTB 42)	110-115	White	Suitable for dry sown conditions, medium tall, moderately resistant to pests and diseases
6	Jayathi (PTB 46)	120-125	White	Resistant to BPH, green leaf hopper, leaf folder, blast and bacterial leaf streak, non-lodging, semi-tall, suitable for all the three seasons
7	Aathira (PTB 51)	120-130	Red, short bold	Semi-tall, non-lodging, moderate resistance to blast and blight diseases and BPH. Suited for I and II crop seasons and also for hilly tracts
8	Aiswarya (PTB 52)	120-125	Red, long bold	Suitable for <i>modan</i> . Resistant to blast and blight diseases. Resistant to BPH. Suited for first and second crop seasons
9	Anashwara (PTB- 58)	125-130	Red, medium bold	Photosensitive, semi-tall variety, suitable only for <i>Mundakan</i> season showing reduced lodging nature. Moderate resistant to blast, sheath blight, leaf folder, stem borer and gall fly
10	Bhadra (MO 4)	120-125 (125-135 for addl. crop)	Red, short bold	Suitable for <i>puncha</i> season in Kuttanad region. Low susceptibility to pests and diseases. Tolerant to BPH. Weakly photosensitive
11	Asha (MO 5)	115-120	Red, medium bold	Suitable for both seasons of Kuttanad. Moderately resistant to pests and diseases. Tolerant to BPH
12	Pavizham (MO 6)	115-120	Red, short bold	Easy to thresh. Fairly resistant to BPH. Moderately resistant to stack burn and sheath rot and fairly resistant to sheath blight
13	Remya (MO 10)	110-120	Red, long bold	Moderately resistant to BPH, gall midge, sheath blight and sheath rot, semi-tall variety. Suitable for all the three seasons. Seed dormancy upto one month
14	Kanakom (MO 11)	120-125	Red, medium bold	Resistant to BPH and moderately resistant to stem borer. Resistant to diseases like rice tungro virus and blast. Moderately resistant to bacterial blight. Semi-tall variety suitable for all the three seasons
15	Ranjini (MO 12)	115-120	Red, medium bold	Dwarf, resistant to blast and BPH
16	Pavithra (MO 13)	115-120	Red, medium bold	Dwarf, medium tillering, resistant to BPH and GM Biotype-5

17	Panchami (MO 14)	115-120	Red, medium bold	Dwarf, medium tillering, resistant to BPH and GM Biotype-5
18	Uma (MO 16)	115-120 Punja 120-135 Virippu	Red, medium bold	Non lodging, resistant to BPH and GM bold Biotype-5. Dormancy upto 3 weeks. Suited to three seasons especially to additional crop season of <i>Kuttanad</i>
19	Karishma (MO 18)	115-120	Red, medium bold	Dwarf, medium tillering, resistant to BPH and medium resistant to GM. Suited to three seasons especially to kari lands of <i>Kuttanad</i> ; tolerant to iron toxicity
20	Gouri (MO 20)	115-120	Red, medium bold	Medium tall, non-lodging, moderately resistant to sheath blight. Suitable for <i>Punja</i> and additional crop seasons of <i>Kuttanad</i> , <i>Mundakan</i> season of <i>Kole</i> lands, first and second crop seasons in double cropped wet lands
21	Vyttila – 1 (Chootupokali-Tall)	115	Red	Suitable for <i>Pokkali</i> areas
22	Vyttila – 2 (Cheruvirippu Tall)	125-130	Red, bold	Suitable for <i>Virippu</i> season in the saline areas of Ernakulam and Alappuzha districts
23	Vyttila – 3	110-115	Red	Suitable for <i>Virippu</i> season in the coastal saline areas of Ernakulam and Alappuzha districts
24	Vyttila – 4	120-125	Red	Suitable for <i>Virippu</i> season in the coastal saline areas (<i>Pokkali</i> ecosystem) and other flood prone areas and deep water situations
25	Vyttila – 5	115-120	Red	Tall, moderately lodging, tolerant to major diseases and pests except BLB, leaf scald, stem borer, leaf folder and rice bug
26	Vyttila – 6	105-110	Red, medium bold	Semi tall, non-lodging, tolerant to salinity, acidity and submergence. Suitable for <i>Virippu</i> season in the coastal saline areas (<i>Pokkali</i> ecosystem)
27	Vyttila-7	110-115	White long slender	Semi tall, non-lodging, high yielding, saline tolerant, suitable for coastal saline ecosystem of Kerala
28	Vyttila-8	115-120	Red, medium bold	Medium tall, non-lodging, high yielding saline tolerant, suitable for coastal saline ecosystem of Kerala

29	Ezhome-2	125-130	Red, medium bold	Suitable for virippu season in the saline coastal Kaipad areas of North Kerala. Non lodging with medium height, non shattering awn-less grains with good cooking qualities. No pest and disease incidence at Kaipad field condition
30	Aarathi (ACV-1)	120-135	Red	Seed dormancy upto one month. Recommended for southern region for <i>Virippu</i> season for delayed sowing and for situations where over-aged seedling may have to be planted. Moderately resistant to sheath blight, sheath rot and BPH
31	Vaishak (PTB 60)	117-125	Red, short bold	Suitable for direct seeding during Kharif season in the uplands. Tolerant to moisture stress, resistant to blue beetle, moderate resistance to stem borer and whorl maggot
Long duration				
1	Lakshmi (Kayamkulam 1)	175-180	Red	Suitable for second crop season. Moderately resistant to leaf roller, blast, blight and sheath blight, fairly resistant to stem borer
2	Dhanya (Kayamkulam 4)	160-165	Red	Photosensitive, moderately resistant to stem borer, gall midge, sheath blight and blast.
3	Rashmi (PTB 44)	150-160	Red	Suitable for growing in the second crop as the mundakan for Koottumundakan system. Resistant to leaf folder and tolerant to gall midge
4	Neeraja (PTB 47)	140-150	White	Moderately resistant to leaf folder, resistant to blast and moderately susceptible to sheath blight, non-lodging, photosensitive, dormant, suited to flood prone and water logged areas and poonthal padams
5	Nila (PTB 48)	160-180	Red, short bold	Photosensitive, capable of producing good grain and straw yields under low fertilizer application. Suitable for Karinkora cultivation. Highly resistant to thrips, BPH and moderately resistant to gall midge, stem borer and sheath blight. Can be cultivated in areas where water supply is assured till the middle of January
6	Mangala Mahsuri (PTB 53)	140-145	Red, medium slender	Multiple resistant, tolerant to iron toxicity and waterlogging. Good response to low fertilizer regimes
7	Karuna (PTB 54)	140-145	Red, long bold	Multiple resistant, tolerant to iron toxicity, good response to low fertilizer regimes, susceptible to brown spot. Specific to II crop season
8	Kottarakara-1 (Cheradi)	140-145	Red	For waterlogged deep soils

9	Makaram (KTR 2)	160-165	Red, short bold	Photosensitive; suitable for eastern lateritic region; II crop component in Kootumundakan system
10	Kumbham (KTR 3)	165-175	Red, short bold	Photosensitive, tolerant to lodging; suitable for eastern lateritic region; II crop component in Kootumundakan system
11	Pankaj (Semi tall)	135-140	White	For ill-drained deep soils, susceptible to leaf blight
12	H4 (Tall)	125-145	Red, bold	For ill-drained deep soils
13	Mahsuri (Tall)	125-145	White, fine rice	Excessive shedding of grains at maturity, susceptible to blast
14	Sagara	180-190	Red	Photosensitive
15	Deepthi	150-160	Red	Semi-tall, photo-insensitive, moderately resistant to blast, leaf folder and stem borer, moderately resistant to drought
16	Ponni	140-145	White, medium slender	Fine grain quality rice
17	White Ponni	135-140	White, medium slender	Fine grain quality rice
18	Ponmani	160-165	White, short bold	Resistant to BPH; high yield potential
19	Pranava	130-135	White, medium slender	Multiple resistant. Suitable for the black cotton soils of Chittoor
20	Swetha (PTB-57)	135-140	White, short bold	Suitable for black cotton soils of Chittoor taluk as a transplanted II crop
21	Dhanu	150-160	Red, short bold	Photosensitive variety suitable for the second crop in Onattukara. Tolerant to sheath blight, brown leaf spot and stem borer
23	Thulam	150	Red, short bold	Semi-tall, photosensitive, non-lodging, non-shattering, tolerant to stem borer, leaf roller and flooding, suitable for eastern lateritic region during second crop season
24	Ezhome-1	135-145	Red, bold medium	Suitable for virippu season in the saline coastal Kaipad areas of North Kerala. Non lodging with large number of sturdy culm of medium height and with purple coloured base. Non shattering awn-less grains with purple coloured apiculus. Good cooking qualities. There is no pest and disease incidence at Kaipad field condition

Tips on quality seed production in rice and maintenance of viability of stored seeds

Ensure that the seeds for further multiplication are either from a research station or government farm or recognized seed producers.

The land used for seed production should be free from volunteer plants of other varieties grown previously.

To ensure genetic purity, a minimum isolation distance of 3 m from other varieties may be given in the field.

Harvesting for seed purpose can also be done leaving a border row of 3 m within the field.

Rogue diseased plants, weeds and off-types in time.

Line planting facilitates roguing and giving alleyways of 30 cm after every 3 m helps in manuring, plant protection operations and supervision.

Drain water at least one week before harvesting to ensure that the plants attain uniform maturity at harvesting. Harvest the crop when 80 per cent of the grains in a panicle are mature (at physiological maturity).

During summer months, ensure that the plants get sufficient water at dough stage till maturity.

Thresh the sheaves on the same day of harvest, as seeds of heaped sheaves may not perform well.

Dry seeds properly and ensure that the moisture content is not more than 13 per cent.

Avoid excess drying in summer months especially for short duration varieties as it reduces the period of viability.

While drying and storing, avoid contamination from yards or through baskets or bags.

Seeds may be stored in damp-proof situations for avoiding absorption of moisture from atmosphere thereby losing viability early.

Polythene bags of 700 gauge or double gunny bags may be preferred for storing.

Polythene bags of 400 gauge density may be preferred for storing paddy seeds dried to 10 per cent moisture content or less.

Never stack seed bags in open floors. Store on pallets or wooden benches. The benches should be 30 cm away from wall and floor for proper aeration.

Never pile more than eight bags in a stack. This should be limited to three bags if the seeds require further drying.

Avoid storing plant protection chemicals, herbicides, fertilizers etc. in seed store.

Fill up the cracks and crevices of storeroom by cementing to make it rat proof.

Spray 2 per cent malathion solution in the godown before storing seeds to check insect pests.

Place pieces of cloth dipped in neem oil between stacked bags or neem oil cake covered in cloth bags inside seed bags to ward-off pests.

Test seed germination at monthly intervals if the seeds are to be stored for more than eight months.

The viability of short duration varieties can be extended for a further period of 2-3 months at 80 per cent level if the seeds are soaked for four hours in water and re-dried in shade, back to original weight at 13 per cent moisture content. Seeds of short duration varieties like Jyothi and Triveni of virippu crop reach this level of germination 9-10 months after harvest and that of mundakan 8-9 months after harvest when stored under ambient conditions.

To keep sprouted pokkali seeds viable for two weeks, the seeds are to be kept in baskets made of plaited coconut leaves lined with koova, banana, karingotta or teak leaves.

Seed rate

Transplanting	60-85 kg ha ⁻¹
Broadcasting	80-100 kg ha ⁻¹
Dibbling	80-90 kg ha ⁻¹

The above seed rates are specified for farmers' field on the basis of minimum germination of 80 per cent. In pokkali cultivation, for Vyttila varieties, 100 kg ha⁻¹ may be sown on the beds or mounds formed in the field.

Seed rate may be enhanced from 80-100 kg ha⁻¹ to 125 kg ha⁻¹ for Kuttanad, provided excess plants are removed in order to maintain optimum plant population.

Seed treatment

Dry seed treatment

Dress seeds with *P. fluorescenes*

@ 10 g/kg of seeds before sowing or with the following fungicides on the previous day of sowing (12 to 16 hours ahead) at dosage given below:

1. Carbendazim 2g per kg of seed

Wet seed treatment

Soak seed for 12 to 16 hours in a solution of *P. fluorescenes* @ 10 g/litre of water

per kg of seed or Carbendazim 2 g / kg of seed per litre of water and drain to induce germination.

Treatment with Carbendazim will protect the seedlings from blast disease up to 30 to 60 days after sowing.

The above seed treatment can also be used for protection from seedling blast in endemic areas.

Soak paddy seeds in CuSO₄ (0.25 per cent) and ZnSO₄ (1 per cent) solution for

24 hours. Drain and keep for sprouting. For soaking 1 kg of seed, 1 litre of micronutrient solution would be needed.

Nursery

For transplanting, healthy seedlings have to be raised in seedbed. Healthy seedlings can cope up better with the field conditions that affect the growth of young rice plants. Adopt wet or dry method for raising seedlings. The choice depends primarily on the availability of water.

Wet method

The wet method can be adopted in areas where water is available as in the second crop season. Seedlings raised by the wet bed method can be harvested one week earlier. The seedbed should be prepared in advance, so that the pre-germinated seeds can be sown in time. As far as possible, fertile lands with irrigation and drainage facilities should be selected for raising the nurseries. Such lands should be suitably located to receive full sunlight. The following are the steps in raising wet nursery.

Plough and harrow the fields two or three times until the soil is thoroughly puddled and levelled. Prepare raised beds 5 to 10cm height 1 to 1.5m width and of convenient length with drainage channels between the beds. The total seedbed area should be 1000 m² for each hectare of the field to be transplanted.

Apply compost or cattle manure @ 1.0 kg/m² of the nursery bed and mix well with the soil at the time of preparation of the field.

Treat the seeds by wet method. Drain and incubate in warm moist place for sprouting. Never allow the seeds to dry up. Moisten them occasionally. Sow germinated seeds on the third day. Delay will result in poor seedling stand.

Irrigation may be commenced on the 5th day after sowing and continued up to the 7th day, to a depth of about 5 cm. After this period, irrigate the seedbed continuously to a depth of about 5 cm in order to control weeds.

Drain occasionally to encourage production of vigorous seedlings with short roots. Flooding the soil with too much water for long periods produces tall and weak seedlings, which do not readily recover during transplanting.

If symptoms of nitrogen deficiency are observed, broadcast urea @ 1 kg for 100 m² as top dressing about 10 days prior to pulling out of seedlings, depending upon the duration of variety.

Dry method

This method is practised in areas where sufficient water is not available and the time of planting is uncertain. During first crop season, wherever transplanting is done depending upon receipt of rainfall, it is safer to adopt this method since growth of the seedlings can be controlled.

Plough the nursery area to a fine tilth. Prepare raised beds of 1 to 1.5 m width, 15 cm in height and of convenient length. Apply compost or cattle manure at the rate of 1 kg/m² of the nursery bed and mix well with the soil at the time of preparation of the field.

Sow the seeds treated as described under dry seed treatment method evenly over the bed and cover with fine sand or soil.

Water the nursery as and when required depending upon the receipt of rains.

Note: Rice seedlings from solarised nursery beds showed high initial growth, early maturity and resistance to leaf blast disease (*ad hoc* recommendation).

Age of seedlings

Seedlings are ready to be pulled out when they attain the stage of 4-5 leaves, about 18 days after sowing in the case of short duration varieties and 20-25 days after sowing in the case of medium duration varieties. Under ill drained conditions, long duration varieties may be planted 30 days after sowing. Seedlings more than 30 days old when transplanted in the field recover slower than younger seedlings, especially, if they suffer stem and root injury. However, during the virippu season, age of seedlings can go up to 35 days in case of medium duration varieties and 25 days for short duration varieties. If the seedlings are over aged, plant at a closer spacing with 3 or 4 seedlings per hill and apply extra dose of nitrogen @ 5 kg/ha as basal dressing.

Irrigate seed beds a day before pulling out the seedlings to soften the soil and to facilitate washing of roots. Wash off mud and soil from the roots carefully and tie the seedlings into bundles of convenient size for transplanting.

Pruning of the top portion and root is not recommended as it inflicts wounds through which disease causing organisms may subsequently enter.

Preparation of land

General

Plough the field thoroughly to incorporate the weeds and straw into the soil. Ensure a smooth, level field for transplanting the seedlings. It would be better to transplant 10-15 days after incorporating organic manure. Before transplanting or sowing, apply manures and fertilizers at the rates specified for the region and varieties as indicated in Table 4. Apply fertilizers on the drained soil at the time of final ploughing and levelling and thoroughly mix into the soil.

Kuttanad

Drain out standing water from the main field. Plough the field thoroughly to incorporate the weeds in the field. Ensure a smooth and levelled field. Maintain a thin film of water to facilitate sowing so that the germinated seeds do not get covered with clayey soil, which affects seedling establishment.

Kole

For the first crop in Kole, after the cessation of the heavy monsoon, dewatering is effected by petti and para or centrifugal pump and rarely by chakkram. Land is ploughed thoroughly and transplanting is done. For the second crop, land is prepared thoroughly and direct sowing of sprouted seeds or transplanting is done.

Onattukara

With the onset of pre-monsoon showers, land is ploughed thoroughly. Dibbling of unsprouted seeds behind the country plough is the common practice.

Pokkali

By April, the bunds are being strengthened and sluices repaired for regulating water level. Fields are then drained during low tide and the sluices are closed. When the soil in the field becomes dry, mounds of 1 m base and 0.5 m height are formed. This facilitates the washing down of the dissolved salts from the surface of the mounds with the onset of monsoon, which are ultimately removed from the field by tidal action. The mounds act as elevated *in situ* nursery and protect the seedlings from flash floods.

A special method is adopted for sprouting the seeds. The seeds are tightly packed in baskets made of plaited coconut leaves, the inside of which is lined by banana or teak leaves. These baskets are then immersed in fresh water ponds for 12 to 15 hours. They are then taken out and stored in shade. The radicle just sprouts and remains quiescent under this condition for more than 30 days. When the soil and weather conditions

become favourable for sowing, the baskets containing the seeds are re-soaked for 3 to 6 hours before sowing. The mounds in the field are then raked and top levelled. The sprouted seeds are sown on the top

of mounds, which act as an in situ nursery. When the seedlings reach a height of 40-45 cm (in 30-35 days), the mounds are cut into pieces with a few seedlings, which are uniformly spread in the field.

Table 4. Fertilizer recommendation for rice, kg/ha

Kind of land / region	Variety	N	P ₂ O ₅	K ₂ O
Uplands (<i>modan</i>)	Upland local varieties	40	20	30
-do-	High yielding short duration varieties	60	30	30
Wetlands (All regions)	-do-	70	35	35
-do-	High yielding medium duration varieties	90	45	45
-do-	Local varieties	40	20	20
-do-	H4	70	45	45
-do-	Mahsuri	50	25	25
Kole lands	*Short duration varieties	90	35	45
	*Medium duration high yielding varieties	110	45	45
Kattukampal and Ponnani kole lands	*Medium duration high yielding varieties	110	45	55
Wetland (Kuttanad)**	Medium duration high yielding varieties	90	45	15
Onattukkara	Dhanya	60	30	30
Koottumundakan	Photo in sensitive varieties for first crop	40	20	20
	Photosensitive varieties for second crop	20	10	10

* Location specific recommendations. Strict surveillance of pests and diseases is a must under such situations.

** Wherever the soil K status is medium to high based on soil test data and also where incorporation of straw is a practice and tidal contribution of the nutrient is significant.

Koottumundakan

In this system of rice cultivation, a mixture of seeds of a non-photosensitive (*virippu*) variety and a photosensitive (*mundakan*) variety of rice in the proportion 70:30 (w/w) is sown during *virippu* season. This system is practiced in areas where sowing / planting of *mundakan* crop is not possible due to excess water in the field. Hence, mixture of the two varieties is sown in the first crop season (April-May). The first crop variety will be ready for harvest in August-September and the second crop variety can be harvested in December-January. No cultivation is practised after the harvest of first crop season variety. But both organic and inorganic manures are applied and incorporated. Though the yield will be less than that of the two independent crops, this type of cultivation is taken up in view of the special circumstances prevailing in such areas.

Transplanting

Transplant seedlings of appropriate age for the variety @ 2-3 seedlings per hill in rows, at spacing as shown in Table 5. Leave wider row of 30 cm after every 3 m to facilitate spraying and other cultural operations. Transplant seedlings at a depth of 3-4 cm

Wet seeding by seed drums and weed control by cono weeder

- Drum seeding and cono weeding can be safely adopted without sacrifice in productivity.
- It is suitable for areas where efficient water management is possible (especially during *rabi*) and during *kharif* in areas where torrential rains are absent (Chittur tract).
- For short duration varieties 15cm spaced and for medium duration varieties 20cm spaced seed drum is recommended.

Manuring

Organic manuring

Apply organic manure in the form of farmyard manure or compost or green leaf @ 5 t ha⁻¹ and incorporate into the soil while ploughing. Vermicompost or coirpith compost @ 2.5 t ha⁻¹ can be substituted for 5 t ha⁻¹ FYM in Onattukara region. The entire quantity of phosphatic fertilizers may be applied along with the organic manures.

Use of biofertilizers is recommended (See chapter on biofertilizers).

Green crop manuring

Cowpea may be raised as an intercrop in dry seeded low land (semi-dry) rice by sowing 12.5 kg/ha seed along with rice to serve as a source of green manure. When the rice field gets submerged with the onset of southwest monsoon, cowpea at the age of about six weeks and at active vegetative stage decays and gets self-incorporated in the soil adding substantial quantity of green manure. Such a system of concurrent growing of cowpea also reduces weed pressure in semi-dry rice.

Concurrent growing of cowpea/daincha A. Dry seeded rice

When there is an undue delay in the onset of monsoon, concurrently grown cowpea in rice fields can be incorporated by spraying 2,4 D @ 1.0 kg ha⁻¹ at 30-40 days after sowing without affecting the yield with a substantial reduction on weed incidence.

B. Wet seeded rice

In wet seeded rice, daincha can be raised as an intercrop by sowing 20 kg seed of daincha ha⁻¹ along with rice (seed rate 60 kg ha⁻¹) to serve as a source of green manure. Daincha can be incorporated by spraying

2, 4D @ 1.0 kg ha⁻¹ at 30 days after sowing thereby adding substantial quantity of green manure. System of concurrent growing of daincha can also reduce the weed pressure in wet seeded rice.

Fertilizer application

The rates of N, P₂O₅ and K₂O recommended as fertilizers are given in Table 4 and stages of application in Table 6.

Table 5. Spacing for rice transplantation

Season	Duration	Spacing	No. of hills/m ²
First crop	Medium	20 cm x 15 cm	33
	Short	15 cm x 10 cm	67
Second crop	Medium	20 cm x 10 cm	50
	Short	15 cm x 10 cm	67
Third crop	Medium	20 cm x 10 cm	50
	Short	15 cm x 10 cm	67

Table 6. Stages of fertilizer application in rice

Kind of land/ region	Variety	Stages of Application									Remarks
		N			P			K			
		BA **	AT	API	BA	AT	API	BA **	AT	API	
Uplands	PTB 28, 29,30	1/3	1/3	1/3	Full			1/2*		1/2	* Full dose as basal is also recommended
	HYshort duration	1/3	1/3	1/3	Full			1/2*		1/2	* Full dose as basal is also recommended
Wet land, direct seeded	General	1/3*	1/3	1/3	Full			1/2*		1/2	*For wet seeded, the first dose to be given 1 week after sowing. For dry seeded, first application to be given after establishment of the seedlings
	Mahsuri	1/3	1/3*	1/3**	Full			1/2		1/2**	*45 DAS, **85 DAS
Wet land, trans-planted	HY short duration	2/3		1/3*	Full			1/2		1/2	*5-7 days before PI
	HY, medium duration	1/2		1/2*	Full			1/2		1/2*	*5-7 days before PI
	Mahsuri	1/2	1/4*	1/4 **	Full			1/2		1/2**	*40 DAP, ** 60 DAP
Onattu-kara	General	1/2	1/4	1/4	Full			1/2		1/2	In very coarse soils, N & K may be given in five equal splits
Wayanad and hilly region	Long duration trans-planted		1/2	1/2		Full*			1/2	1/2	*Along with first application of N & K
	Direct seeded		1/2*	1/2		Full**			1/2*	1/2	*45 days after seeding, **With first application of N & K
Pokkali	General	Full*			Full*						*Apply entire quantity at the time of dismantling of mounds
Kootu-mundakan	First crop	1/2		1/2	Full			1/2		1/2	
	Second crop	Full*			Full*			Full*			*Entire quantity as single dose immediately after the harvest of first crop

BA=Basal application; AT = At tillering; PI = At panicle initiation; HY = High yielding; DAP = Days after planting; DAS = Days after seeding

For *modan* cultivation (upland crop) and direct seeded crop in wet lands, apply nitrogen in three equal split doses, first as basal dressing, second at tillering stage (three weeks after seeding) and the third at panicle initiation stage (about thirty days before flowering). Apply the full dose of phosphorus at the time of land preparation as basal dressing. Apply potash either in a single dose as basal or in two split doses half as basal and half at the panicle initiation stage.

In Kuttanad region, wherever wet broadcasting (direct seeding) is adopted, give the first basal application of the nitrogen at the time of letting in water after drying the field. Water-soluble phosphorus can be recommended for application in two split doses in Kuttanad region, as basal and at maximum tillering stage.

The general principle to be followed is that in light soils as well as in soils with high leaching, nitrogen may be applied in three or four split doses according to the duration of the variety.

For typical Onattukara region, where soil is sandy loam and with iron toxicity problem, apply 5 tonnes of organic matter or vermicompost 2.5 t and 67.5 kg K₂O ha⁻¹.

During the first crop season, when basal application of nitrogen is not possible due to incessant rains, basal dose can be shifted to 15 days after transplanting.

In coarse sandy loam soils with high percolation as in Onattukara region, nitrogen and potash fertilizers may be applied in five equal splits at planting, 15th, 38th, 53rd and 70th day for medium duration varieties

which coincide with the stages of early tillering, neck node differentiation, early reduction division and heading stages respectively in the case of medium duration varieties.

For Thiruvananthapuram and Malappuram districts, phosphorus application is essential for increasing rice yields. Rock phosphate may be substituted for superphosphate.

Basal dose of nitrogen may be postponed to initial tillering phase of rice crop, especially during the rainy season. Split application of potash @ 50 per cent basal, 25 per cent at tillering and 25 per cent at panicle initiation stage is recommended for this region.

In sequential cropping of rice, application of 50 per cent of the nutrient requirement (on nitrogen equivalent basis) as organics (FYM, rice straw, green manure) and 50 per cent as fertilizers during kharif season and the entire dose of nutrients as fertilizers in rabi season enhances the grain and straw yield. The organics may be incorporated 3 weeks before transplanting.

Application of 25 per cent of the nutrient requirement as organics and 75 per cent as fertilizers during kharif season and reducing the fertilizer dose of rabi by 25 per cent gives comparable yield with full POP recommendation during both seasons.

Specific dose of fertilizer for Koottumundakan system

A fertilizer dose of $N:P_2O_5:K_2O$ 20:10:10 kg ha⁻¹ for the first crop and 30:15:15 kg ha⁻¹ for the second crop is recommended for high yield in *Koottumundakan* system. N and K₂O may be applied in two equal parts for the first crop, one as basal and other at panicle initiation stage. P₂O₅ may be applied fully as basal. The fertilizer for second crop may be applied as a single dose immediately after the harvest of the first crop (*ad hoc* recommendation)

Fertilizers ($N:P_2O_5:K_2O$) @ 40:20:20 kg ha⁻¹ for virippu and 20:10:10 kg ha⁻¹ for the photosensitive mundakan crop are recommended as economic dose for the northern region under *Koottumundakan* practice.

Methods of fertilizer application

For pre-planting application, apply the fertilizers at the final ploughing. In areas where availability of water is assured, temporarily draining the field one day prior to application and re-flooding after twelve hours is recommended for top dressing of fertilizer.

For increasing the efficiency of urea for top dressing, mix urea with six times its weight of slightly moist soil and apply to the field 24-28 hours after mixing. Oil seed cakes such as *punna* and *neem* cakes can also be mixed with urea (1 part of oil cake + 5 parts of urea by weight) for increasing fertilizer use efficiency. This method is particularly useful for basal application of nitrogen. Under special conditions of drought and water-logging, apply nitrogen as foliar spray. Urea may be applied as a low volume spray at 15 per cent concentration using power sprayer or at 5 per cent concentration using a high volume sprayer, the quantity applied in one application being limited to 15 kg ha⁻¹.

When zinc deficiency is noticed, apply zinc sulphate @ 20 kg ha⁻¹. Early stages of zinc deficiency are evidenced by interveinal chlorosis, bleaching of midribs and light yellow colouration of the leaf. Older leaves develop brown rusty spots and are extremely brittle. Zinc sulphate and potash should not be applied on the same day.

The effect of zinc application can persist upto five years depending on the soil and cropping pattern. Hence soil application is not required for every season. Soil zinc status should be monitored before application to avoid accumulating toxic concentration of zinc.

Split application of water-soluble phosphatic fertilizers in two equal splits as basal and at maximum tillering stage is effective in giving higher grain and straw yield than the full dose as basal dressing under certain situations.

Application of chemical fertilizers can be dispensed for Vyttila-1 and Vyttila-2 in *pokkali* rice fields.

When the soil has less than 10 ppm of calcium chloride extractable sulphur or 15 ppm of phosphate extractable sulphur, substitute urea with ammonium phosphate sulphate to correct sulphur deficiency. For medium duration rice grown in brown hydromorphic soils ammonium phosphate sulphate may be used to supply 25 kg sulphur per hectare to protect the crop from sulphur deficiency (*ad hoc* recommendations).

Application of magnesium as basal dose in the form of magnesium sulphate (16 per cent MgO) or magnesite (40 per cent MgO) or dolomite (10 per cent MgO) @ 20 kg MgO/ha is effective in giving a significant increase in grain and straw yield of rice in magnesium deficient soils. In non-deficient soils, a marginal increase in grain and straw yield is also obtained. On per unit MgO basis, magnesite is more concentrated and cheaper than magnesium sulphate.

In iron toxic laterite soils of Kerala, application of 120 kg potash ha⁻¹ + lime 150 kg ha⁻¹ + silica 100 kg ha⁻¹ (as sodium silicate 250 kg ha⁻¹ or fine silica 100 kg ha⁻¹ or rice husk ash 500 kg ha⁻¹) is recommended for higher yields.

Liming

In general, addition of lime is absolutely necessary when the pH is lower than 5.5 and it is advisable when pH varies between 5.5 and 6.5.

Apply lime @ 600 kg ha⁻¹ in two split doses, the first dose of 350 kg ha⁻¹ as basal dressing at the time of first ploughing and the second dose of 250 kg ha⁻¹ as top dressing about one month after sowing/transplanting.

For *pokkali* areas, apply lime @ 1000 kg ha⁻¹, 50 per cent at the time of preparation of mounds and the rest at the time of dismantling the mounds.

A time lag of one week should be given between application of lime and fertilizers. For top dressing, lime may be applied one week prior to the application of fertilizers.

Water management

Maintain water level at about 1.5 cm during transplanting. Thereafter increase it gradually to about 5 cm until maximum tillering stage. Drain water 13 days before harvest.

Note: In areas where water for irrigation is assured and where acidity is high, draining and reflooding every 15 days is recommended. In flood prone areas, aged seedlings of Mahsuri or other varieties recommended for waterlogged conditions may be planted. The date of planting may be adjusted so as to avoid synchronization of the critical stages of maximum tillering or heading with the usual flood period in the tract.

During the *mundakan* crop season, water level of 5 cm need not be maintained continuously after the cessation of northeast monsoon. Five centimetre irrigation once in 6 days will be quite adequate for project areas where water is assured.

For summer rice (in situation where the ground water level is low, i.e., within 1m from the surface), 5 cm irrigation two days after disappearance of ponded water is sufficient instead of 5 cm continuous submergence throughout the crop period.

Irrigation schedule for rice under limited water resources

For summer rice under limited resources of water, phasic stress irrigation can be practised to the advantage of saving substantial quantity of irrigation water without any significant reduction in yield. About 20-30 per cent more area can be irrigated with the same water resources by adopting any of the following phasic stress irrigation schedules (Table 7). Depending up on the schedule, water saving ranges from 24-36 per cent of the requirement for 5 cm continuous submergence throughout the crop growth. Grain yield reduction in the above practice is only 0.1 per cent to 1.6 per cent.

Weeds and their management

Common weeds in rice fields of Kerala are:

Grasses: *Oryza rufipogon* (varinellu), *Echinochloa crusgalli* (kavada), *E. colona* (kavada), *E. stagnina* (kavada), *Sacciolepis interrupta* (polla), *Isachne miliacea* (chovverippullu, naringa).

Sedges: *Cyperus iria* (manjakora, chengoal), *C. difformis* (thalekkattan), *Fimbristylis miliacea* (mungai)

Table 7. Irrigation schedule for rice under limited water resources

Schedule	Stages		
	Rooting to max. tillering	Max. tillering to heading	Heading to maturity
Category I	Continuous submergence	Saturation point*	Saturation point*
Category II	Saturation point*	Continuous submergence	Continuous submergence
Category III	Continuous submergence	Continuous submergence	Hair cracking of surface*
Category IV	Hair cracking of surface*	Continuous submergence	Hair cracking of surface*

* Irrigation at 5 cm to be given at the stages marked.

Broad leaved weeds: *Monochoria vaginalis* (neelolppalam), *Ludwigia perennis* (neer-grampu), *Limnocharis flava* (nagappola), *Ammania baccifera* (nellicheera).

Ferns: *Salvinia molesta* (African payal), *Marsilea quadrifolia* (naalilakodian), *Azolla pinnata* (azola).

Algae: *Chara* spp. (chandi), *Spirogyra* spp. (payal).

Control

Keep the rice fields free from weeds up to 45 days either by hand weeding or by use of herbicides. The recommendation for use of herbicides in different systems of rice culture are given below:

A. Dry seeded rice [upland and lowland (semi-dry) rice]

Spray any of the following pre-emergent herbicides: butachlor @ 1.25 kg ai ha⁻¹, oxyfluorfen @ 0.15 kg ai ha⁻¹, pendimethalin @ 1.50 kg ai ha⁻¹, pretilachlor @ 0.75 kg ai ha⁻¹ on the same day of seeding or within six days of seeding.

B. Wet seeded rice (direct seeding with sprouted seeds under puddled conditions)

Spray any of the following herbicides:

(a) Butachlor @ 1.25 kg ai ha⁻¹ 6-9 days after sowing;

(b) Pretilachlor + safener @ 0.45 kg ai ha⁻¹ 3-5 days after sowing. Give a follow up application of 2,4-D @ 0.8 kg ai ha⁻¹ at 20 days after sowing.

(c) Pretilachlor @ 0.45 kg ai ha⁻¹ at 3-4 DAS and one light hand weeding at 28 DAS for wet sown rice in kole land.

To control *Echinochloa* sp. spray cyhalofop butyl @ 0.08 kg ai ha⁻¹ at 15-18 days after sowing.

In certain areas like Kuttanad, wild rice has become a menace in recent years. If effective water control is possible, the following agronomic practices can successfully control the infestation of wild rice.

(a) Coat dry seeds with 20 per cent calcium peroxide using 4 per cent PVA solution as an adhesive and then broadcast in the field with 10-15 cm column of standing water.

(b) Maintain the water level for 10-12 days to prevent germination of wild rice. (c) Drain the field and apply N and K fertilizer as per recommendation for rapid growth of the rice seedlings. (d) Where yeranda (common teal) is a problem, bird scaring should be arranged till the water is drained.

C. Transplanted rice

Apply any of the following herbicides at 0-6 days after transplanting.

Pendimethalin @ 1.5 kg ai ha⁻¹, butachlor @ 1.25 kg ai ha⁻¹. For controlling broad leaved weeds and sedges 2, 4-D sodium salt may be applied @ 1 kg ai ha⁻¹ on 25 DAT.

Pre emergence application of 2, 4-D sodium @ 0.8 kg ai ha⁻¹ on 5 DAT can control all the three groups of weeds viz. grasses, sedges and broad leaved weeds.

Wherever fields are levelled and water management could be effectively done 2, 4-D can be mixed with 10kg urea/ha and broadcast on 20 DAS/DAT. This would save the spraying charges.

Apply 2,4-D Na salt as pre-emergence broad-spectrum herbicide @ 0.8 kg ai ha⁻¹ at 5 days after transplanting. Spray on moist soil surface with a fan nozzle to give blanket coverage. Use 300-400 litres of spray fluid.

Almix 20WP(Chlorimuron ethyl 10% +Metsulfuron methyl 10% @ 4g ai ha⁻¹ + 0.2% surfactant at 20-25 DAS/DAT is equally effective as 2,4-D for controlling sedges and broad leaved weeds in rice fields. This herbicide is effective for controlling *Marsilia quadrifolia* also. Almix can be given as follow-up application to cyhalofop butyl used for the control of grass weeds with a minimum of one-day gap between the applications.

Table 8. Cropping pattern / system

General	Rice	Rice	Fallow
	Rice	Rice	Vegetable/pulses/oil seeds/ green manure
Kuttanad	Water fallow	Water fallow	Rice
	Rice	Rice	Water fallow
Onattukara	Rice	Rice	Oil seed/green manure
Kole / Kaippad	Water fallow	Rice	Water fallow
	Water fallow	Rice	Rice
	Water fallow	Water fallow	Rice
Pokkali	Rice	Fish / prawn	—
High ranges	Rice	Rice	Fallow

Management of *Salvinia molesta* (African payal)

Trampling salvinia *in situ* in the wet lands a week before transplanting will control the weed and add to soil fertility. Herbicides should be applied only in areas where protected drinking water supply is available (see Annexure).

Precaution while using herbicides

(1) Apply herbicides at the recommended dose and time. (2) Drain the field before herbicide application. (3) Re-flood after 48 hours to prevent further weed germination when post emergent herbicides are used. For pre-emergent herbicides, wait for a week before re-flooding the field. (4) Use herbicide nozzle (flood jet / flood fan) for herbicide application. (5) Move at uniform speed when applying herbicides. (6) Spray without gaps and overlapping. (7) Use 300-400 litres of water per hectare for spraying the herbicide.

Pests and Diseases

Adopt control measures only if the pest / disease population exceeds the economic threshold levels which are given in Table 9.

Pests

Rice stem borer

(*Scirpophaga incertulas*)

Symptoms

In the vegetative phase, the central shoot dies off turning yellow in colour (dead heart). In the ear bearing stage, the ear head appears completely chaffy and white in colour (white ear head). Both come out easily when pulled up and show indication of feeding injuries at the base.

Management

1. Collect egg masses from the nursery plants and observe for parasitisation.
2. Cultivate tolerant varieties like IR-20 in endemic areas.
3. In areas where stem borer occurs as a serious pest in all seasons, apply any one of the following insecticides first 15-20 days after transplantation and then at the boot leaf stage keeping minimum water level: Cartap hydrochloride (spray or granules), quinalphos, carbaryl, carbosulfan, flubendiamide, indoxacarb, malathion or spinosad.
4. Use sex pheromone for the control of rice stem borer as detailed in Table 10.

Gall midge (*Orseolia oryzae*)**Symptoms**

Presence of silver shoot in the place of central leaf is the prominent symptom. The symptom appears from the nursery to the flowering stage. However, in very young seedling the silver shoots are not always expressed. Instead, a swelling at the basal portion and excess tillering are often noticed.

Management

1. Use tolerant varieties like Pavithra, Panchami and Uma
2. Avoid late transplantation during the first crop season.
3. Careful monitoring of the crop seasons in the month of July during additional crop season and October during puncha season.
4. Use optimum seed rate of 100 kg ha⁻¹
5. Destruction of collateral host like wild rice, *Cynodon dactylon*, *Ischaemum aristatum*, *Echinochloa* spp. and *Isachne* sp.

Table 9. Economic threshold levels (ETLs)

Crop stage and pest	Economic threshold levels
<i>A. Nursery</i>	
1. Green leaf hopper	1-2 insects/m ²
2. Gall midge	1 silver shoot (gall)/ m ²
3. Stem borer	1 moth or 1 egg mass/ m ²
4. Blast	5% disease severity
<i>B. Planting to pre-tillering</i>	
1. Leaf folder	2 freshly damaged leaves/hill
2. Yellow stem borer	10% dead hearts or one egg mass or one moth / m ²
3. Gall midge	1 gall/ m ² in endemic areas or 5% affected tillers in non-endemic areas.
4. Brown plant hopper	5 to 10 insects/hill
5. Green leaf hopper	10 insects/hill (in RTV endemic areas 21 insects/hill)
6. White backed plant hopper	10 insects/hill
7. Rice Hispa	2 adults or 2 damaged leaves/hill
<i>C. Mid tillering</i>	
1. Leaf folder	2 freshly damaged leaves/hill
2. Stem borer	10% dead heart or 1 moth or 1 egg mass/ m ²
3. Gall midge	10% silver shoots
4. Brown plant hopper	10 insects/hill
5. Green leaf hopper	10-20 insects/hill
6. Hispa	2 adults or 2 damaged leaves/hill
7. Blast	Light (5-10% disease severity)
8. Bacterial blight	Light (2-5% disease severity)
9. Sheath blight	10% or more affected tillers
10. Tungro	1 affected hill/ m ²
<i>D. Panicle initiation to booting</i>	
1. Stem borer	1 egg mass or 1 moth/ m ²
2. Leaf folder	2 freshly damaged leaves/hill
3. Green leaf hopper	20 insects/hill
4. Brown plant hopper	15 to 20 insects/hill
5. White backed plant hopper	15 to 20 insects/hill
6. Blast	5 to 10% leaf area damaged
7. Bacterial blight	Light to moderate (2-5% disease severity)
8. Sheath blight	10% or more tillers affected
<i>E. Flowering and after</i>	
1. Brown plant hopper	25 to 30 insects/hill
2. Climbing cutworm	4-5 larvae/ m ²
3. Rice bug	1 or 2 bugs/hill
4. Blast	5% leaf area damaged or 1-2% neck infection
5. Sheath rot / brown spot / slight panicle discolouration	2-5% tillers affected
6. Sheath blight	10% or more tillers affected
7. Stem borer	2% white ear head

Source : *Manual on Integrated Pest Management in Rice*, Directorate of Plant Protection, Quarantine and Storage

Note: (1) The population should be estimated on the basis of careful and regular surveillance.

(2) When natural enemies of brown plant hopper, green leaf hopper, stem borers and leaf folders are present, application of chemical measures can be delayed or dispensed with.

**Table 10. Sex pheromone used for the control of rice yellow stem borer
(ad hoc recommendation)**

Chemical name of the pheromone	2-(z)-9-hexadecanol 2-(z)-11-hexadecemol in 1:3 identified from female moths
Sex attracted	Male moths
Description of the trap	Sleeve trap with pheromone loaded rubber septa
Uses	Pest monitoring : Three traps can be fixed in a triangular fashion at about 80 m apart Mating confusion
Number of traps required/ha	Twenty
Sources of availability	Entomology Department, Directorate of Rice Research, Rajendra Nagar, Hyderabad.
	Pest Control India limited, Bio-control Research Laboratories P.O. Box. 6426, Yelahanka P.O., Bengaluru-560 064 Karnataka
	Ecomax Agrosystems 302, Faigha Plaza Basher Bagh, Hyderabad.
	Dr.David Hall and Dr.Allen Cork, Natural Resources Institute, Kent ME4, 4TB, United Kingdom.
	The Managing Director, Som IPM System (India) Ltd., Plot No.101, 1 Floor, Srinagar Colony, Hyderabad.

6. Dipping germinated seed in 0.2 per cent chlorpyrifos solution for 3 h before sowing give protection up to 30 days.

7. In transplanted crop the root of seedlings may be dipped in 0.02 per cent chlorpyrifos suspension for 12 h prior to planting.

8. The nursery treatment has to be followed by main field treatment, 10-15 days after transplantation using any one of the following insecticides: quinalphos, carbaryl.

9. In areas where the pest is of regular occurrence, apply granules of quinalphos (250g ai/ha) or chlorpyrifos 10G

(0.5 kg ai/ha) within 10 days after sowing. The granules should be broadcast in 2-3 cm of water and the field should be impounded for at least 4 days.

Rice bug (Leptocoris acuta)

Symptoms

Look for the presence of bug in the field during the early ear bearing stage. Due to de-sapping, grains show brownish discoloured patches on the husk.

Management

1. Strict vigilance is necessary at milky stage.

2. Keep the field and bunds free of weeds and grasses.

3. Avoid overlapping cultivation in an Ela.

4. When the bug is seen in large numbers apply one of the following insecticides: Malathion, carbaryl.

Note: Since the occurrence of the bug coincides with the flowering stage, application of the insecticide may be done either before 9 a.m. or after

3 p.m. so that fertilization of the flowers is not adversely affected.

Leaf folder (Cnaphalocrocis medinalis)

Symptoms

The leaves of the plant are seen folded, rolled and often webbed together with white patches on them indicating the areas fed by caterpillar. When such folded leaves are opened up, larvae can be seen. Shaded conditions and application of excess nitrogen are conducive for leaf folder attack.

Management

1. Open up the leaf folds with the help of a thorny twig.

2. Apply one of the following insecticides in the field where the symptoms of attack are manifested: quinalphos, carbaryl, phosalone, acephate, flubendiamide, cartap hydrochloride, indoxacarb and dichlorvos.

Note: In the initial stages restrict spraying to infested patches only. The field may be sprayed completely in case the infestation occurs uniformly.

Brown plant hopper (*Nilaparvata lugens*)

Symptoms

Yellowish circular patches appear here and there in field. The plants in these areas dry up very soon (hopper burn) and the yellowing and drying extend rapidly. Examine the plants as soon as the yellowing appears. Presence of the hoppers at the base of the plants confirms the infestation. Very close planting leads to enhanced attack.

Management

1. Use resistant varieties such as Jyothi, Bharathy, Aiswarya, Kanakom, Nila etc. for cultivation.
2. Apply one of the following insecticides as soon as the yellowing symptom is observed, covering the infested patches and the areas surrounding the patches: carbaryl, quinalphos, acephate, thiamethoxam, phosalone and imida-clopid. While spraying, care has to be taken to see that the insecticides reach the base of the plants.
3. Drain away water from the field and keep it in that conditions until the pest population dwindle.
4. In Kuttanad tract, early planting of paddy in September-October is advisable, wherever possible.
5. Allow alleways after every 3m rows
6. Avoid spraying synthetic pyrethroids.

Rice case worm (*Nymphula depunctalis*)

Symptoms

Leaves of plants are eaten by the caterpillars, which remain within small cylindrical cases and are seen hanging on the leaves. It occurs in ill drained fields. The cases may be seen floating on water also.

Management

1. Drain away the water from the fields.
2. If infestation is severe, apply carbaryl dust or spray.

Rice swarming caterpillar

(*Spodoptera mauritia*)

Symptoms

It appears in the field sporadically and cyclically in large swarms and feed on crops gregariously. The nursery and early stages of the crop are attacked leaving the plant as mere stumps.

Management

Apply carbaryl or trichlorfon as soon as the caterpillars are noticed.

Rice hispa (*Dicladispa armigera*)

Symptoms

The adults feed on the green tissues of the leaves and the feeding scars appear as short white lines on the leaf surface. The grubs mine the leaves causing formation of white blotches. Early stages of the crop are more susceptible.

Management

Spray any contact insecticide.

Rice thrips (*Stenchaetothrips biformis*)

Symptoms

The crop is highly susceptible in the nursery stage for the first 23-25 days after transplanting/sowing. The tips of leaves get rolled longitudinally into needle like outgrowths and turn whitish. In severe cases, the lower leaves also turn yellowish. The infestation may be rated as mild, if there is less than three needle like leaves and as severe, if there are more than three outgrowths with the lower leaves also showing chlorosis and scorching.

Management

In severe infestations, apply carbaryl 625 g ai/ha + DDVP 100 per cent EC/AF 250 ml/ha or dimethoate or quinalphos or phenthoate (EC/AF formulation).

Whorl maggots (*Hydrellia philippina*)

Infestation is common in the nurseries and in the main fields up to six weeks after transplantation. Yellowish patches and streaks are seen along the margins of leaves, which may become deformed.

Leaf hoppers (*Nephotettix* spp.)

Symptoms

General yellowing of the leaves is seen, if the attack is severe. When the plants are disturbed, the jassids are seen jumping out.

Management

Apply carbaryl, quinalphos, imidacloprid or flubendiamide if needed.

Rice mealy bug (Brevinnia rehi)**Symptoms**

Weak yellowish stunted plants are seen in patches. White waxy fluff is seen in leaf sheaths.

Management

Dimethoate 0.05 per cent is effective in controlling the pest.

Rice root nematode (Hirschmanniella oryzae)**Symptoms**

Infests paddy roots and make them partially hollow. Feeding adversely affects absorption of water and nutrients. Plants show stunted growth in patches. Tiller production is affected.

Management

Dip the roots of seedlings in 0.2 per cent dimethoate for six hours before transplanting in tracts where nematode attack is detected.

Rice cyst nematode (Heterodera oryzae)

The cyst nematode occurs in various proportions in certain areas of the State. The symptoms of infestation include leaf chlorosis, stunting and reduction in the number of leaves, earhead length etc. In seriously affected patches, yield is substantially reduced.

Table 11. Insecticide guide for rice pest control

Sl No.	Insecticide	*Dosage	Insect controlled
1	2	3	4
1.	Acephate	800g of 75 SP per ha	Rice leaf folder, Brown plant hopper
2.	Carbaryl	20kg of 10 DP per ha	Rice case worm, brown plant hopper
3.	Carbaryl	2kg of 50 WP per ha	Rice leaf folder, rice bug, gall midge, rice swarming caterpillar and brown plant hopper
4.	Carbosulfan	17 kg of 6 G per ha	Rice stem borer
5.	Cartap hydro-chloride	25kg of 4 G per ha	Rice stem borer and leaf folder.
6.	Cartap hydro-chloride	1kg of 50 SP per ha	Rice stem borer and leaf folder.
7.	Chlorpyrifos	0.2% suspension	Germinated seed dip for 3 hrs. against gall midge
		0.02% suspension	Seedling root dip for 12 hrs. against gallmidge
8.	DDVP+ carbaryl for combined application	250 ml of DDVP 100 EC/AF and 1.25kg carbaryl 50 WP per ha	Brown plant hopper and rice thrips
9.	Dichlorovos or DDVP	500ml of 100 EC/AF per ha	Rice leaf folder
10.	Dimethoate	0.2% suspension 0.05% suspension	For seedling dip against rice nematode For foliar application against thrips
11.	Flubendiamide	125 g of 20 WDG per ha	Rice stem borer, whorl maggot and leaf folder
12.	Flubendiamide	50 ml of 480 SC per ha	Rice stem borer, whorl maggot and leaf folder

1	2	3	4
13.	Imidacloprid	150ml of 200SL per ha	Brown plant hopper
14.	Indoxacarb	200ml of 15.8 EC per ha	Rice stem borer, whorl maggot, leaf folder and blue beetle.
15.	Malathion	1000ml of 50 EC/ AF per ha	Rice stem borer, whorl maggot and rice bug
17.	Phosalone	1000ml of 35EC/AF per ha	Brown plant hopper
18.	Quinalphos	1000ml of 25 EC / AF per ha	Rice stem borer, brown plant hopper, gall midge, rice leaf folder and rice jassids
19.	Quinalphos	25kg of 5 G per ha	Gall midge and rice stem borer
20.	Spinosad	100 ml of 45 SC per ha	Rice stem borer, whorl maggot and leaf folder
21.	Thiamethoxam	100 g of 25WG per ha	Brown plant hopper and safe to green mirid bug (BPH Predator)
22.	Trichlorfon	1000-1500ml of 50 EC/ AF per ha	Rice swarming caterpillar

*

Dosage applies to the crop stage of booting and beyond. For early stages the quantity of insecticides can be reduced

** This is a special micronised formulation with better spreading quality
 SP = Soluble Powder;
 DP = Dustable Powder;
 G = Granule;
 EC = Emulsifiable Concentrate;

AF =
Aqua
Flowable;
WP =
Wettable
Powder;
S =
Sprayable
suspension
SL =
Soluble
Liquid
SC =
Suspension
Concentrate;
WDG =
Water
Dispersible
Granule.

Note :

1. Sub-lethal concentration of insecticides may lead to pest resurgence
2. Granular application is recommended only upto the booting stage
3. Spot application method should be resorted to wherever possible
4. Use 200, 300 and 500 l/ha of spray fluid at 10, 25, 45 DAT or 30, 45, 60 DAS respectively for high volume spray equipment
5. When low volume spray equipment is used the spray fluid can be limited to 90, 120, 180 l/ha, but the quantity of insecticide should remain the same as used in high volume spray 200, 300 and 500 l/ha respectively
6. Waiting periods for quinalphos and malathion are 7 and 3 days respectively
7. In regions where BPH is a regular pest avoid the application of the following insecticides due to the chances of resurgence: deltamethrin, carbaryl, and quinalphos.

Diseases

Fungal diseases

Blast (*Pyricularia grisea*)

Major symptoms of the disease are leaf blast and neck blast. Sometimes nodal infection is also seen. Symptoms on leaves appear as spindle shaped water soaked greyish green spots, which gradually enlarge in size and develop to spots with grey centre and brown margin. Such spots coalesce together resulting in drying up of the leaves and collapse of the entire plant. The fungus infects the neck of the panicle causing dark brown to black lesions resulting in rotten neck or neck blast. This will also lead to breaking of the panicle at the neck region. The grains will be partially filled or unfilled. If the panicle is infected before grain filling stage, the entire panicle will be chaffy.

Sheath blight (*Rhizoctonia solani*)

Symptoms first appear on the lower leaf sheath near the water level as oval, oblong or irregular greenish grey spots, which enlarge and become greyish white with brown margin. Under favourable conditions, the disease spreads to the leaves also. Irregular greenish grey lesions with dark brown margins develop on leaves. Under humid conditions, white mycelial growth of the fungus as well as initially white and later brown sclerotia of the fungus loosely attached to the affected portion are also seen.

Brown spot (*Helminthosporium oryzae*)

On the leaves small definite spots of oval or oblong shape and dark brown colour are formed. On susceptible varieties, spots are larger in size and are having light brown or grey center with dark reddish brown margin.

The symptoms also appear on glumes as black oval or oblong spots or whole surface of the grain turns black and velvety.

Narrow brown spot (*Cercospora oryzae*)

Numerous narrow linear short brown spots are formed on the leaves.

Stack burn (*Alternaria padwickii*)

Symptoms appear on leaves as large oval or circular dark brown lesions with narrow distinct margins. On the affected grains pale brown to whitish spots with dark brown margin bearing black dots in the center

are formed.

Leaf scald (*Rhynchosporium oryzae*)

Lesions with typical zonations with dark coloured wavy lines usually start from the tip of the leaves and extend downwards or arise from the margin and advance inside.

Sheath rot (*Sarocladium oryzae*)

The fungus infects the leaf sheath enclosing the panicle causing oblong or irregular brown spots which later develop into a lesion with dark brown margin and grey center. The young panicle remains inside the sheath or rather chocked and emerge only partially. The grains become chaffy. A white powdery growth of the fungus can be seen on the surface of rotten sheath particularly inside the sheath.

False smut (*Ustilaginoidea vireus*)

The symptoms visible only after flowering when infected grains get transformed to yellow to orange spore balls, which later turns to dark green or black.

Udbatta (*Ephelis oryzae*)

Whole panicle transformed into a cylindrical rod covered with white mycelia. Later they become hard bearing many black dots.

Foot rot (*Fusarium moniliformae*)

Drying of leaves and leaf sheath discolouration of lower nodes and adventitious root formation are the major symptoms. Plants turn pale yellowish green, thin and some times show abnormal elongation or rot in patches in the field.

Bacterial diseases

Bacterial blight (*Xanthomonas oryzae* pv. *oryzae*)

Bacterial blight is characterised by two phases of infection. *Kresek* and leaf blight. *Kresek* is drying or wilting of the whole plant if it affects the crop in early stages of growth in nurseries and upto 3-4 weeks after transplanting. The infected leaves become grayish green and begin to roll along the midrib and dry. In severe cases the affected hills may be completely killed.

Leaf blight symptom starts as water soaked lesions on the tip of the leaves and increases in length downwards along the margins. Initially the lesions are pale green in colour and later turn into yellow to straw coloured stripes with wavy margins. Occasionally the linear lesions may develop anywhere on the leaf lamina or along the midrib with or without marginal stripes. Lesions may cover the entire leaf blade. The bacterial blight disease can be confirmed by ooze test. Cut the affected plants at the base and dip it in a glass of water and hold it against the light for few minutes without shaking. From the cut ends white milky bacterial ooze will come out as streams.

Bacterial leaf streak

(*Xanthomonas oryzae* pv. *oryzicola*)

Narrow brownish yellow translucent interveinal lines joining to form large dirty white patches.

Note :

1. Spray fresh cowdung extract for the control of bacterial blight. Dissolve 20 g cowdung in one litre of water; allow to settle and sieve. Use supernatant liquid.
2. Application of bleaching powder @ 5 kg ha⁻¹ in the irrigation water is recommended for checking the spread of bacterial leaf blight particularly in the *kresek* stage.
3. Before application of antibiotics for the control of bacterial blight, identify the disease by observing the bacterial ooze.
4. For control of sheath blight and sheath rot the following prophylactic measures may be adopted.
 - a) Apply neem cake-coated urea as recommended under fertilizer application.
 - b) Apply 50 per cent more potash than normal recommended dosage in split application.
 - c) Control weeds as suggested under weed control.
5. For control of sheath rot, spray the fungicides at the time of panicle emergence.
6. In organic farming, for the management of sheath blight of rice, spraying Azadirachtin 0.03 per cent EC or Azadirachtin 0.15 per cent w/w @ 2.5 litres ha⁻¹ at the time of first disease appearance.
7. Spraying of any fungicide, preferably dithiocarbamate before the heading stage of the crop may be followed for the control of false smut and leaf scald diseases. Wherever control measures are adopted for sheath blight and sheath rot diseases, separate treatments are not needed for the control of leaf scald and false smut. Use 500 litres of water for high volume spraying and 200 litres for low volume spraying. It is advisable to use chemical on a rotational basis than using the same product continuously.

Table 12. Guide on control of rice diseases

<i>Foliar Sprays</i>			
A. Non systemic			
Sl. No.	Chemical	Dosage	Disease controlled
1.	Zineb	2 kg ha ⁻¹	Blast, brown leaf spot, narrow leaf spot and sheath blight
2.	Mancozeb	2 kg ha ⁻¹	Leaf spot diseases but more generally used against brown leaf spot
3.	Thiophanate	500 g ha ⁻¹	Sheath blight and blast
B. Systemic			
1.	Iprophenphos 48 EC	500ml ha ⁻¹	Blast and sheath blight
2.	Carbendazim	500 g ha ⁻¹	Sheath blight and sheath rot
3.	Carboxin	500g ha ⁻¹	Sheath blight and sheath rot
4.	Hexaconazole 5EC	800-1000 ml ha ⁻¹	Sheath blight
5.	Propiconazole 25EC	500 ml ha ⁻¹	Sheath blight
6.	Carpropamid 27.8 SC	500 ml ha ⁻¹	Blast
7.	Isoprothiolane 40 EC	750ml ha ⁻¹	Blast
C. Antibiotics			
1.	Aureofungin sol	60 g ha ⁻¹	Blast and brown leaf spot
2.	Streptocycline	15 g / 300 l ha ⁻¹	Bacterial blight
3.	Validamycin 3% liquid	1000 ml/500 l ha ⁻¹	Sheath blight
D. Biocontrol Apply <i>Pseudomonas fluorescens</i> for the control of fungal and bacterial diseases (Adhoc recommendation; See chapter on biocontrol agents).			
E. Bio Pesticides for Sheath blight control : Spray Neem formulations Azadirachtin 0.15 per cent w/w@2.5 l ha ⁻¹ or Azadirachtin 0.03 per cent EC @ 2.5 l ha ⁻¹ at the time of Sheath blight disease occurrence			

Hints for reducing cost of cultivation

1. Grow a green manure crop like daincha in April-May in areas where the virippu crop is usually transplanted.
2. If azolla is available, this may be applied instead of green leaf or FYM at 5t ha⁻¹.
3. Puddle and level the field thoroughly. This will help to reduce the cost of weed control and also the loss of water and nutrients through percolation.
4. Maintain optimum plant density per unit area, i.e., 50 hills/m² for medium duration varieties and 67 hills/m² for early duration varieties.
5. Plant the seedlings shallow (3-4.5 cm), as shallow planting increases the tillering of seedlings.
6. Control the weeds during the vegetative phase itself. One weeding thoroughly on the 30th day after sowing is ideal. Use herbicides for weed control, wherever it is cheaper than hand weeding.
7. Under good management practices (vide item 1-6) reduce the dose of fertilizer N to half the present recommended level. Apply the fertilizer when the plant is able to make the best use of it, at tillering and seven days before panicle initiation.

8. When the field preparation and planting are done on rainy seasons, postpone the application of the basal dose of nitrogen to the early tillering stage (10 days after planting).
9. *Use of Leaf Colour Chart (LCC)*: The leaf colour chart (LCC) is an instant, easy and low cost technique for N diagnosis of current crop and N topdressing in rice. The uppermost fully opened leaf of the primary tiller has to be taken for observation. Ten hills are selected at random for an area of 400 sqm and thus 10 leaves form the sample size. The leaf colour is compared with LCC and the readings are taken. The average of 10 readings gives the LCC value of the plot. Observation is taken at 10 days interval from 20 days after transplanting or 25 days after sowing upto heading. If the threshold value is less than 4, N@ 25-30 kg/ha is recommended for application. If the inherent soil N is low, basal N @ 25-30 kg ha⁻¹ is to be applied.
10. When the amount of available N is limited, apply it 7 days before panicle initiation. This is the best time for top dressing N.
11. Adopt agronomic practices for increasing fertilizer use efficiency such as: (a) Incorporating ammoniacal N in the reduced zone in the soil (b) Incubating urea with moist soil (1:6) for 24 hours (c) Blending urea with neem cake.
12. Choose fertilizer materials, which are cheaper, e.g. urea is cheaper than ammonium sulphate; rock phosphate is cheaper than superphosphate.
13. Apply phosphatic and potash fertilizers once in two seasons in areas where there is no marked response for these nutrients.
14. Adopt integrated pest management against insect pests and diseases.
15. Harvest the crop at optimum moisture content in order to avoid loss due to shedding and also for improving the recovery of rice.

Mechanical Transplanting in Rice

A. Mat Nursery Preparation

1. Use very thin polythene sheet
2. Select nursery area near to the main field
3. Cow dung powder may be incorporated @ one third volume of the puddled soil and spread over the sheet at a thickness of 10-15mm
4. The sheet may be spread after leveling the nursery area
5. Form small bunds along the boundaries of the polythene sheet, after spreading it, to impound water.
6. Mat nursery may be of 900mm width and convenient length
7. Just sprouted seeds (4th day of soaking) may be uniformly spread over the mat area @ 0.4 to 0.6kg m⁻²
8. Seedbed may be mulched, preferably using green leaves
9. Sprinkle water over the seedbed for four days twice daily and keep soil in saturated condition. Never allow mat to dry up.
10. On 4th day remove the mulch and keep standing water to a height of 3/4th of seedling height. Keep this condition till the end of nursery period.
11. Closely observe the nursery for pest and disease attack and adopt control measures, if required.
12. When seedlings reach a height of 150mm, mat is ready for transplanting.
13. Drain the required mat area 6-12 hours before cutting for feeding into the transplanter trays.
14. Cut the mat strips 225mm wide and 450mm long to match the measurements of the transplanter trays.
15. Do not allow nursery to over grow. Over growing will lead to entanglement of seedlings with fingers of transplanter and clogging of fingers due to thick and hard mat of roots.

B. Main field

1. Prepare the main field adequately, without clods and stubbles interfering with functioning of finger.
2. Prepare main field sufficiently early to avoid floating of puddle while transplanter is working
3. Final preparation may be done on the day of transplanting in case of sandy loam soil, but may be done 3-4 days prior to planting in case of clayey soils, depending upon settling time of the puddle to consolidate.
4. A thin film of water only is needed in the main field at the time of transplanting.
(Too much water in the field will lead to floating of seedlings and too dry condition to non-anchoring of seedlings).
5. Main field should have irrigation and drainage facility
6. Wetting of mat may be required while transplanter is in operation.

Integrated pest management (IPM) in rice

The indiscriminate use of pesticides for pest control has led to disturbances in natural ecosystem leading to resurgence of pests, secondary pests outbreak, toxic hazards and residues besides environmental pollution. This has led to major emphasis on integrated pest management and gained greater momentum. Integrated pest management programmes promote favourable, ecological, economic and sociological outcome, which is accomplished by the best mix of pest control tactics. The use of appropriate scouting tactics, proper diagnosis of pest, the use of economic thresholds and conservation of naturally occurring

biocontrol agents are fundamental components of a sound Integrated pest management programme. The use of chemical is restricted. It is used only if it is absolutely essential based on surveillance. The important components in the IPM are the use of tolerant/resistant varieties, regulating planting density, adjusting the time of planting / sowing, group farming practices, cultural management of pests, integrated nutrient management, removal of weeds, use of botanical pesticides and preservation of natural enemies. For the preservation of natural enemies, collect the egg masses of pests in perforated polythene bags and keep them in the field so that the parasites that emerge can establish in the field effectively.

Trichogramma chilonis and *Tricho-gramma japonicum* are egg parasitoids which effectively control egg mass of leaf roller, stem borer, skippers and cutworms. The parasitoids have to be released 15-30 days after transplantation or 25-30 days after sowing or immediately after noticing moth activity in the field. The release rate is 1 lakh parasitoids/ha of both size (5cc ha⁻¹). The release has to be carried out at weekly intervals. The trichocard have to be cut into small pieces (minimum 10 pieces) and released in main field, 6-8 releases is necessary to control the pest.

Precaution : If larval attack is observed in the field, necessary organic/inorganic insecticides have to be used and a gap of 7 days has to be given before next release. The trichocards have to be placed during early morning or late evening hours and should not come in direct contact with sunlight.

Table 13. Common natural enemies of insect pests found in rice ecosystem

A. Predators	Description of biocontrol agents	
	Characters	Pest attacked
1	2	3
Spiders 1. Wolf spider 2. Lynx spider 3. Jumping spider 4. Long jawed spider 5. Dwarf spider 6. Orb spider	Female lays 200-800 eggs in life-time of 3-4 months. Both nymphs and adults are voracious feeders. It consumes 5-15 insect pests in a day depending up on the size	Leaf and plant hoppers, leaf feeding caterpillars and adult stem borers.
Damsel and dragon flies	Multi-coloured with transparent narrow wings. Nymphs are aquatic and can climb up rice stems to search food. Adults fly normally below the rice canopy in search of flying insects	Stem borers, hoppers and other flying insects.
Mirid bugs	Adults are green and black at the shoulder. Nymphs are greenish. They can consume 7-10 eggs or 1-5 hoppers a day	Leaf hoppers and plant hoppers
Water bugs	Broad shouldered adults can be either winged or wingless. Each female lays 20-30 eggs in rice stumps above the water level. Life span 1-2 months. Winged adults disburse when rice plants dry up.	Plant hopper and other small soft-bodied insects
Water treaders	They are solitary feeders	Stem borer larvae and hoppers that fall on water surface

1	2	3
Water striders	Each strider takes 5-10 prey daily. They live for 1-1.5 month; lay 10-30 eggs	Rice hoppers, moth and larvae that fall on the water surface
Ground beetle	Hard bodied insects. Larvae are shiny black and adults are reddish brown. Active predators, which pupate in the soil. Consume 3-5 larvae /day	Plant hoppers and larvae of leaf folders
Rove beetle	7mm long with short elytra and blue tip of abdomen. Found on rice plant, water and ground surface; active during night.	Leaf hoppers, plant hoppers and larvae of leaf folder and hairy caterpillar.
Lady bird beetles	Active during daytime. Found in the upper half of rice canopy. Feed on small and slow moving prey as well as on exposed eggs. Grubs are more voracious than adults and consume 5-10 prey. Produce 150-200 offsprings in 6-10 weeks	Plant hoppers
Crickets, Sword tailed crickets	Egg predators	Eggs of stripped borer, leaf folder, armyworm and nymph of plant hoppers and leaf hoppers.
Grasshoppers, Meadow grass hoppers	Distinguished from true grasshoppers by its long antennae, which are more than twice as long as its body length. Active at night and abundant in older fields. Can consume 3-4 yellow stem borer egg masses per day	Eggs of rice bug and stem borer, nymphs of plant hoppers and leaf hoppers
B. Parasitoids		
1. Egg parasitoids		
<i>Trichogramma japonicum</i> <i>Telenomus chilonis</i> <i>Tetrastychus</i> sp.	Dark coloured tiny insects. Female wasp lays 20-40 eggs in the host egg. Development from egg to adult stage takes 10-40 days.	Stem borers and leaf folders
<i>Gonatocerus</i> sp. <i>Anagris</i> sp.	Brown to dark yellowish brown tiny wasp. Females can reproduce without mating. Adults live 6-7 days and parasitise on an average 8 eggs/day	Plant hoppers and leaf hoppers
2. Larval parasitoids		
<i>Cotesia</i> sp. <i>Stenobracon</i> sp. <i>Macrocentrus</i> sp. <i>Xanthopimpla</i> sp. <i>Charops</i> sp.	Dark coloured wasp. Lives for 4-7 days	Larvae feeding on stem, leaf and other parts of rice plant

C. Birds

- Install artificial nesting sites viz., wooden nest boxes in the rice fields (@ 12 nests ha⁻¹) for colonization by cavity nesting birds (maggie robin, common myna, barn owl, etc) which play a vital role in managing a variety of insect and non-insect pests. The nest boxes should be installed at a height more than 8 feet on wooden pole, tree trunk, etc.
- Installation of bird perches @ 50 ha⁻¹ increases the activity of insectivorous birds and reduces the pest population in rice fields. The perches should have a minimum height of 75 cm above the crop canopy for better visibility for the birds.
- Tying metalised reflective ribbon @ 20 - 25 rolls ha⁻¹ at 75 cm above the canopy of the vulnerable stage of the crop on bamboo poles, 2.5 3.0 m apart at convenient length, twisted at every 1 m length

scares away the depredatory birds.

Effective and eco-friendly pest management in wet land rice ecosystem

1. Cultivate tolerant varieties
2. Monitor the field at least at weekly intervals
3. Pest and natural enemy population should be monitored for deciding spraying schedule.
4. Spraying should be avoided during the reproductive phase.
5. Spot application of insecticide may be adopted in heavily infested pockets to control further spread of the pest and to conserve the existing natural enemy population in rice ecosystem.

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MAIZE (*Zea mays*)

Maize can be grown throughout the year at altitude ranging from sea level to about 300 m. Maize grows best in areas with rainfall of 600-900 mm. It requires fertile, well-drained soil with a pH ranging from 5.5-8.0, but pH 6.0-7.0 is optimum.

Season

As a rainfed crop, maize is grown in June-July or August-September. The irrigated crop is raised in January-February.

Varieties

Hybrids: Ganga Hybrid-1, Ganga Hybrid-101, Deccan hybrid, Renjit, Hi-Starch.
Composite varieties: Kissan Composite, Amber, Vijay, Vikram, Sona, Jawahar.

Seeds and sowing

Seed rate: 20 kg ha⁻¹

Plough the land three times and prepare ridges and furrows. Dibble one seed per hole at a spacing of 60 cm x 23 cm for the rainy season crop. For irrigated crop, beds are prepared. Here, seeds are sown in lines and earthed up later in to small ridges to form furrows when the crop reaches knee height.

Manuring

Apply FYM/ compost @ 25 t ha⁻¹ at the time of preparation of land. The recommended fertilizer dose is 135 kg nitrogen, 65 kg phosphorus and 15 kg potash per ha. Apply full dose of phosphorus and potash and 1/3 dose of nitrogen as basal. Apply 1/3 nitrogen, 30-40 days and the rest 60-70 days after sowing.

Aftercultivation

Hand hoeing and weeding on the 21st and 45th day after sowing.

Irrigation

Irrigate the crop on the day of sowing and on third day. Subsequent irrigations may be given at 10-15 days intervals.

Plant protection

Need based application of carbaryl is recommended for control of pests

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SORGHUM (*Sorghum bicolor*)

Sorghum is a plant of hot and warm localities. The optimum temperature for growth is 30°C and it needs about 250-400 mm rainfall. Excess moisture and prolonged drought are harmful. It is fairly tolerant to alkalinity and salinity.

Seasons

Rainfed crop : May-August

Irrigated crop : January-April

Varieties

Co.1, Co-10, Co-12, Co-17, K-1, K-2

Hybrids - CSH-1 to CSH-4, Co-11, Co-1

Seeds and sowing

Seeds are sown @ 12-15 kg ha⁻¹. Dibble two seeds/hole, at a spacing of 45 x 15 cm.

Manuring

For both irrigated and rainfed crops FYM / compost may be applied @ 5 t ha⁻¹.

Fertilizers may be applied as follows:

Nutrients / ha	Irrigated crop	Rainfed crop
N	90 kg	45 kg
P ₂ O ₅	45 kg	25 kg
K ₂ O	45 kg	25 kg

Apply FYM and entire quantity of phosphorus and potash as basal dose. Apply nitrogen in two equal splits, half as basal and the rest 30 days after sowing.

Aftercultivation

Thinning, weeding and hoeing may be done on the 20th day after sowing.

Irrigation

Irrigate the crop on the day of sowing and thereafter at 10 days interval.

Plant protection

Apply carbaryl at milky stages for the control of earhead bugs.

Spray zineb @ 2kg ha⁻¹ on 30th and 45th day after germination for the control of foliar diseases.

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RAGI (*Eleusine coracana*)

Ragi is suited for cultivation in areas with annual rainfall of 700-1200 mm. It does not tolerate heavy rainfall and requires a dry spell at the time of grain ripening. It grows well in altitudes of 1000-2000 m with average temperature of 27°C. Ragi is cultivated mostly in red lateritic soils. Relatively fertile and well drained soils are the most suitable.

Seasons

Ragi is not a season bound crop and hence can be cultivated throughout the year, if moisture is available. It is usually grown during the following seasons.

Main season: June-September

Late season: July-October

Summer: Dec-Jan to March-April

Varieties

PR-202, K-2, Co-2, Co-7, Co-8, Co-9, Co-10

Seeds and sowing

Seed rate

Direct sown crop : 5 kg ha⁻¹

Transplanted crop : 4-5 kg ha⁻¹

Spacing : 25 x 15 cm

Prepare the nursery field to a fine tilth. Incorporate FYM/compost 5 t ha⁻¹ and form beds and channels. Sow seeds uniformly on the beds and cover by stirring the soil. Apply carbaryl 10 per cent on the edges of the beds against ants. One week before uprooting seedlings, apply ammonium sulphate @ of 1 kg per 100 m². A nursery area of 480 m² is necessary to raise seedlings for one hectare. Transplant the seedlings when they are three weeks old.

Manuring

Plough the field 3-4 times and incorporate FYM or compost 5 t ha⁻¹. Apply nitrogen, phosphorus and potash @ 22.5 kg ha⁻¹ each before sowing or planting. Topdress nitrogen @22.5 kg ha⁻¹ 21 days after sowing or planting.

Aftercultivation

Weeding should be done three weeks after sowing and completed before top dressing.

Irrigation

Irrigate the field on the day of transplantation. Irrigation at weekly intervals increases growth rate and yield.

Plant protection

For controlling grasshoppers, spray carbaryl 50 per cent WP @1.2 kg ha⁻¹. The pink stem borer, which causes dead heart symptoms, is often serious in the early stages of the crop. Insecticidal control is not needed for mild to moderate levels of infestation.

For controlling blast, spray mancozeb 750-1000 g ha⁻¹.

Harvesting

Harvest the crop when the ears are yellowish brown. Heap the harvested ears and cover the heaps with ragi straw for 2-3 days before threshing.

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Citation:

Kerala Agricultural University. 2011. *Package of Practices Recommendations: Crops*. 14th Edition. Kerala Agricultural University, Thrissur. 360p.